

Semi-total Domination in Unit Disk Graphs

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1. Problem Studied

- The paper studies the **Semi-total Dominating Set (STDS)** problem in Unit Disk Graphs (UDGs).
- A set $D_{t2} \subseteq V$ is a STDS if:
 1. D_{t2} is a dominating set.
 2. For every $u \in D_{t2}$, there exists $v \in D_{t2}$ such that $\text{dist}(u, v) \leq 2$.

2. Motivation and Applications

- STDS is a natural variant lying between classical dominating sets and total dominating sets.
- Relevant in wireless networks and sensor placements where both coverage and minimal connectivity are required.

3. Main Contributions

- Proved that the STDS problem is **NP-complete** in UDGs via a reduction from Vertex Cover on planar graphs.
- Proposed a **6-factor approximation algorithm** for STDS in UDGs.
- Introduced a $2 + \ln(D+1)$ approximation for general graphs, improving over the previous bound of $2 + 3\ln(D+1)$.
- Showed that a PTAS for STDS in UDGs exists with an approximation factor of $2 + \varepsilon$.

4. Techniques Used

- **NP-Completeness:** Used grid embedding and geometric gadget constructions to simulate vertex cover behavior.
- **Approximation Algorithm:**
 1. Compute a maximal independent set D .
 2. For each vertex in D lacking a neighbor within distance 2 in D , add a neighbor from $N(u)$ to set T .
 3. Return $D \cup T$ as the semi-total dominating set.

5. Results Summary

Graph Type	Approach	Approx. Factor	Time Complexity
UDG	Greedy + Local Fix	6	$O(nk)$
UDG (PTAS)	Local Search + Packing	$2 + \varepsilon$	$n^{O(1/\varepsilon \log \varepsilon)}$
General Graph	Set Cover-Based	$2 + \ln(D + 1)$	—

6. Comparison with Prior Work

- Improves over the $2 + 3 \ln(D + 1)$ result from Henning and Pandey (2019).
- The first work to give a dedicated algorithm and complexity result for STDS in UDGs.
- Relates to: $\gamma(G) \leq \gamma_{t2}(G) \leq \gamma_t(G)$.

7. Limitations / Assumptions

- Assumes no isolated vertices.
- Requires geometric embedding (disk centers must be known).
- Maximal independent set construction may not be efficient for very large or distributed settings.

8. Observations and Future Ideas

- Potential to explore distributed or dynamic versions of the 6-approximation algorithm.
- LP relaxation and rounding might yield tighter bounds.
- STDS in other geometric graphs or higher dimensions could be investigated.